Same Sentences, Different Order: What the Arrangement of Training Text Teaches Small Transformers

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Abstract

Training pipelines decide how text is arranged when they pack, order or generate it, but the effect of arrangement is rarely tested with content held fixed. We trained small transformers on artificial languages in which the same sentences about 32 kinds of object came in different orders. We then described one object of a new kind and measured whether the models expected a second object of that kind to keep its shape more than its colour. Trained on random order, they expected the shape to change; trained on the same sentences grouped by kind, they expected it to stay. Across conditions their expectations tracked what the next sentence usually does, though not the expectation of change in random order, and two things mattered: a feature fixed within kinds, and nearby sentences about different objects rather than the same object again. These results use test nouns the models had never seen. With nouns from training whose kind always had one shape, a shape given in the context was not carried to the next mention; when a kind's shape changed from one pass to the next, the models carried it forward, but only after the same noun. Models trained on one short fixed file, each position read 160 to 290 times, memorised it; the grouping effect was then about half as large, and the random-order index went from reliably negative to slightly positive. In these small models, what is placed near what in training data, not only what is included, decides what they generalise.

1 Introduction

Every training pipeline arranges its text. It packs documents into windows, orders them and, more and more, generates synthetic text whose layout someone chooses (Shi et al., 2024; Zhao et al., 2024). Work on training data mostly asks about content (Warstadt et al., 2023; Feng et al., 2024). Arrangement alone, with content held fixed, is rarely tested, although natural text differs in it: a word that has appeared once in a document is far more likely to appear again soon than its frequency predicts (Church, 2000), and talk to toddlers returns to the same objects in clusters (Slone et al., 2023). In transformers, burstiness of this kind decides whether in-context learning emerges (Chan et al., 2022). We ask a smaller question. With the same sentences, what does arrangement teach a model to expect about the next mention of a kind, and which properties of an arrangement matter?

We need a regularity that holds across items, so that arrangement can make it easier or harder to see. We borrow one from word learning: things with the same name tend to share a shape (Landau et al., 1988; Kemp et al., 2007). We give a model one sentence about an object of a new kind, *A snerg is a krannish pulek skorli thing*, where the three adjectives name a shape, a colour and a texture. We then ask how much more it expects a second snerg to keep the shape than the colour. We use this only as a probe of what transformers learn from their data; our claims are about transformers, not children.

A model can meet this probe in two ways. It can carry features forward from the previous sentence, whatever its noun, or it can use what it has stored about a noun. We separate the two by testing with novel nouns, about which nothing is stored, and with nouns from training.

We trained 510 small transformers (3.4M to 25.6M parameters) on artificial languages whose content and order we control. Our main models saw every pass of their corpus arranged afresh; others were trained, as is common, on one fixed file. We find:

1. **Arrangement teaches what the next sentence does.** With novel nouns (tokens the models had never seen), random order taught the models to expect a shape to change and grouping by kind taught them to expect it to stay. Across

conditions their expectations tracked a simple learner that predicts each feature from the previous sentence, although it does not explain the expectation of change.

2. **Two things mattered.** The gain needed a feature fixed within kinds, and nearby sentences about different objects of a kind taught it more selectively than the same object described again.

3. **Trained nouns took up the context's shape only when their kind's shape changed.** With a noun from training whose kind always had one shape, a shape given in the context was not carried to the next mention (exploratory). When a kind's shape changed between passes, the models carried it forward, but only after the same noun.

4. **Heavy repetition changes the result.** Models trained on one fixed file, each position read 160 to 290 times, memorised it; the grouping effect was about half as large, and the random-order index was no longer negative.

We also measured the same arrangement statistic in natural corpora and probed pretrained models (Section 5). The confirmatory tests were registered in stages; some registrations were written after earlier results were known, and Appendix A gives the dated record.

2 Related work

Arrangement and order. Packing related documents into one window improves how pretrained models use their context (Shi et al., 2024; Zhao et al., 2024). Curricula that order examples by difficulty have given mixed results (Bengio et al., 2009; Rohde and Plaut, 1999; Diehl Martinez et al., 2023). At child scale, the local order of child-directed speech affected what models learned, while its global order by age did not (Feng et al., 2024). We hold content fixed and vary only local arrangement.

In-context and in-weights learning. Chan et al. (2022) found that in-context learning emerged when items of a class appeared in bursts, more strongly when item meanings were dynamic rather than fixed, and traded off against learning stored in the weights; Singh et al. (2023) showed that it can fade as the stored solution takes over, and Reddy (2024) traced its onset to induction heads. We study the same trade-off in next-word prediction over text, with the content held fixed and only the arrangement varied.

Artificial languages and repeated data. Artificial languages let a model's preferences be traced to its input (McCoy et al., 2020; White and Cotterell, 2021; Kallini et al., 2024). Such studies typically reuse one generated corpus over many passes, from about two to a few dozen in these examples (McCoy et al., 2020; Kallini et al., 2024; Murty et al., 2023). The number of passes matters: training long on fixed data changed hierarchical generalisation (Murty et al., 2023); at a fixed budget, small sets of repeated examples beat larger sets of single-use examples on synthetic maths tasks (Charton and Kempe, 2024); and at scale up to about four epochs of repeated data changed loss negligibly (Muennighoff et al., 2023). The BabyLM Challenge capped training at ten epochs in 2025, judging tens or hundreds of repeated exposures developmentally implausible (Charpentier et al., 2025).

Shape bias in networks. Networks trained on images or artificial objects can show a shape bias (Ritter et al., 2017; Feinman and Lake, 2018), and a hierarchical Bayesian model learns it from how kinds vary (Kemp et al., 2007). We use it as a probe, not as a target.

3 Method

3.1 Corpora and arrangements

Each corpus describes 32 kinds of object, each named by a novel noun (*wug*, *dax*, . . .), with 14 exemplars per kind. Each exemplar has a shape, a colour and a texture, each a novel word from a list of ten, and is described by one sentence in one of five frames (Figure 1; Appendix I). Shape is named in all five frames, colour in three and texture in one. The frames are the same in every corpus, so every conclusion rests on comparisons between corpora. The 448 sentences of a corpus (1,664 for Objects), each used once, make up one pass. Table 1 defines the corpora, the arrangements and the measures.

Arrangement. In random order every pass is shuffled. In grouped order the sentences about each kind are cut into runs of four and the runs are shuffled across kinds; every sentence occurs as often as in random order. Graded orders keep each run with a set probability and otherwise split it, giving intermediate clustering. We measure clustering with one statistic, B, used for natural corpora too: among consecutive mentions of object nouns no more than 100 words apart, the share that name the same kind, divided by the same share after the

Term	Meaning
Regular	Each kind keeps one shape; colour and texture vary.
No stable feature	Shape drawn at random per exemplar; nothing fixed within a kind.
Colour stable	Each kind keeps one colour instead (fixed files only).
Objects	13 objects per kind, each described four times.
Reassigned	Each kind gets a new shape in every pass and keeps it within that pass.
Random order	Every pass shuffled.
Grouped order	Sentences about a kind cut into runs of four; runs shuffled across kinds.
Same- / different-object run	Four descriptions of one object / of four objects of one kind.
B	Clustering: share of nearby noun mentions that name the same kind, relative to shuffled text (random 0.9, grouped 23).
Fresh / fixed text	Every pass arranged anew, no window read twice / one fixed file, each position read 160 to 290 times.
Δ	Preference (nats) for a continuation that repeats a feature of the context sentence.
Index	$\Delta_{\text{shape}} - \Delta_{\text{colour}}$.

Table 1: Terms used in this paper.

sentences of each document (here, each pass) are shuffled. $B = 1$ means no clustering beyond what the content predicts.

Fresh and fixed text. Our main models were trained on fresh text: 7,500 passes of a corpus (2,100 for Objects), each arranged with its own random draw, 34 to 35 million tokens in all, read in non-overlapping 128-token windows so that no window was read twice. A second set of models was trained on one fixed file of 11,648 sentences (about 120,000 tokens) with windows at random positions, so that each position was read 160 to 290 times. Colour stable corpora, three of the six levels of clustering and several controls exist only as fixed files (Table I1, Appendix I).

3.2 Models and test

We trained decoder-only transformers of 3.4, 11.0 and 25.6 million parameters (4 to 8 layers) for 5,000 to 10,000 steps, with five seeds per condition at each size; a seed sets the corpus, the weights and the batch order (Appendix C). Every artificial word is one token.

Each test item gives the model one frame-1 sentence about an exemplar, then two continuations that differ in one word: one repeats the probed feature and one does not (Figure 1). Δ is the difference in summed log probability between them. Trained models generally prefer words they have just seen, so we report the index, $\Delta_{\text{shape}} - \Delta_{\text{colour}}$, which cancels copying that affects both features equally, together with Δ for each feature. With one exemplar, nothing in the context shows which feature is fixed, so differences between conditions come from training. Untrained models give an index of 0.02 nats.

Test nouns are of two kinds. **Novel nouns** never occur in training. Their input embeddings were never trained, because a tie between input and output embeddings was lost on the training hardware, and the models showed no sign of recognising a repeated novel noun (Appendix E). **Trained nouns** occur in training. To keep items comparable, an item never offers a trained noun its own kind's fixed feature, so in Regular corpora both continuations give the noun a shape it never had (Appendix F). Each item is also given with a different noun in the continuations; the **noun-specific index** is the index after the same noun minus the index after a different noun.

Statistics. Models of three sizes with the same seed share a corpus, so the five seeds are the units of replication. We average sizes within each seed and compare conditions by paired t tests over the five seed means (4 degrees of freedom); estimates and 95% intervals in the text are these seed-level values. Registered contrasts were one-sided with Holm (1979) correction within each registration. For fresh-text models the registered test was the seed-level test; for fixed-file models it was a linear mixed model or a Welch test over models, with seed-level results beside them in Appendix E. Results given with a Holm p were registered; others are exploratory. Every contrast reported as reliable had the same direction at all three model sizes. Table 3 lists the main registered tests and marks those written after related results were known.

4 Results

4.1 Arrangement teaches what the next sentence does

With novel nouns, Regular models trained in random order came to expect the shape to *change*: their index was −1.56 nats [−2.45, −0.67], below zero in all 15 models (Table 2). The same sentences in grouped order gave 9.52, higher in all five seeds and at every size (difference 11.08 [9.57, 12.59], Holm $p < .001$). At $B = 3.5$, near the clustering of object labels in speech to children, the index

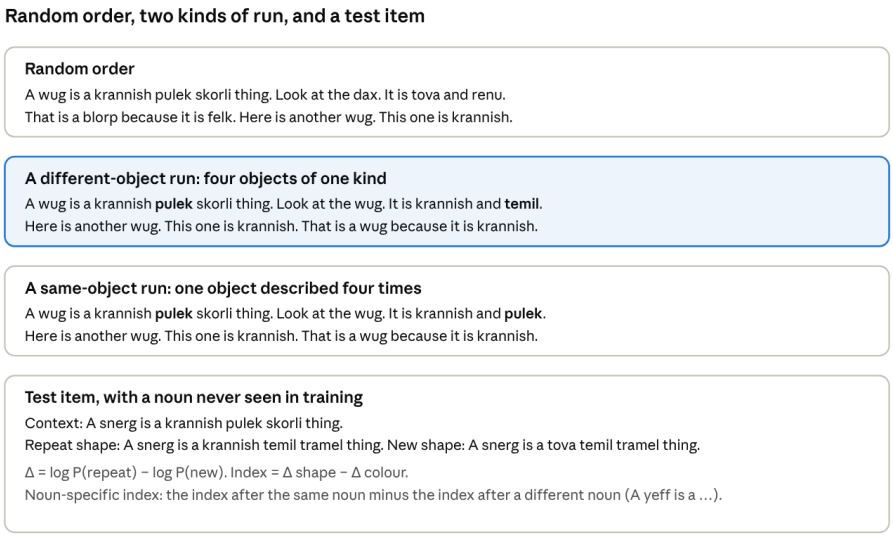


Figure 1: Random order, the two kinds of run, and a test item. In the different-object run the colour changes from one wug to the next; in the same-object run it repeats.

		Novel nouns				Trained nouns, Δ shape	
Corpus	Order (B)	Δ shape	Δ colour	Index	Shape over texture	Same noun	Different noun
Regular	Random (0.9)	−1.55	0.01	−1.56	−1.44	0.03	0.04
Regular	Graded (3.5)	1.18	−0.10	1.28	1.21	0.06	0.04
Regular	Grouped (23)	6.88	−2.64	9.52	6.56	0.06	−0.08
No stable feature	Grouped (23)	−3.88	0.10	−3.98	−4.32	−3.58	−1.18
Objects	Same-object runs (24)	6.87	0.99	5.87	6.73	−0.09	−0.28
Objects	Different-object runs (24)	8.05	0.46	7.60	7.91	0.04	−0.32
Reassigned	Random (0.9)	0.22	0.02	0.21	0.39	0.24	0.24
Reassigned	Grouped (23)	6.81	−2.99	9.80	5.62	10.01	0.08

Table 2: Models trained on fresh text, after one exemplar (nats; 15 models per row). Left: novel test nouns, same noun in the continuation. Right: preference for repeating the context's shape with trained test nouns, after the same or a different noun (exploratory). Appendix E gives standard deviations, model sizes and every fixed-file condition.

was 1.28, above random order (2.84 [1.41, 4.27], Holm $p = .005$), but small and variable across sizes (1.37, 2.27 and 0.21).

These expectations follow the corpora. In grouped order the next sentence is about the same kind 72% of the time and then always repeats the shape; in random order it is about the same kind 3% of the time. A learner that predicts each feature from the previous sentence alone, ignoring the noun, gives a reference index, and the models tracked it across conditions (Pearson $r = .88$ over the eight fresh-text conditions, .95 over 12 fixed-file conditions; exploratory and descriptive, since conditions share seeds, and driven mainly by grouped against ungrouped orders; Appendix E). The models also learned how long runs last: with two or four sentences about the new kind in the context, the grouped Regular index fell from 9.52 to 7.88 and 4.24, because after four sentences about a kind the next is about the same kind only 3% of the time.

The gain needed a fixed feature. When nothing was fixed within kinds, grouped models expected the shape to change, at −3.98 (difference from grouped Regular 13.50 [11.48, 15.52], Holm $p < .001$). Two results come only from fixed-file models. When colour was fixed instead, the preference reversed: grouped Colour stable models favoured colour, at −3.20, below fixed-file No stable feature models at −0.44 (difference 2.76 [1.99, 3.52], Holm $p < .001$). Across six levels of clustering the index rose with $\log B$ (slope 1.63 [1.46, 1.80] nats per log unit, Holm $p < .001$; Figure 2).

The reference does not explain two values: it is near zero for random order (−0.01) and for No stable feature corpora (0.02), where the models gave −1.56 and −3.98. We have no tested account

of these expectations of change. Both arise when the test noun carries no stored shape (Section 4.3).

4.2 Different objects, not the same object again

At matched clustering, runs of different objects gave a larger index than runs describing one object four times: 7.60 against 5.87 (difference 1.72 [0.54, 2.90], Holm $p = .008$; higher in all five seeds; fixed-file models agreed, 5.65 against 2.99). The components show why. After same-object runs the models expected colour to repeat more (Δ colour 0.99 against 0.46), as it does in those runs. Their index stayed positive because they also repeated shape, which the frames name more often, more than the corpus warrants; part of this may be copying. Runs of different objects show what varies within a kind and what does not, and taught the fixed feature more selectively. The noun-blind reference predicts this direction, with a larger gap (0.39 against 3.43; Appendix E), so the contrast is what the corpora warrant rather than a further property of the models.

4.3 Trained nouns and the context

Trained nouns behaved differently (Table 2, right; exploratory). In every fresh-text corpus where a kind keeps one shape throughout (Regular and Objects), models tested with a trained noun showed almost no preference about shape in any arrangement: Δ shape was −0.09 to 0.06 after the same noun, against −1.55 to 8.05 with novel nouns (grouped Regular: 0.06 [−0.27, 0.39] against 6.88 [5.22, 8.54]). So a shape given in the context, never the noun's own, was not carried to the next mention. This is consistent with the models predicting a trained kind's shape from its noun, but our items cannot test that, because they never offer the stored shape. Arrangement still shaped what the noun does not predict: in grouped Regular corpora, trained-noun items showed an expectation that colour would change (Δ colour −1.94 [−2.87, −1.02]). In No stable feature corpora, trained nouns, like novel ones, gave a negative Δ shape after the same noun (−3.58 against −3.88), but much less after a different noun (−1.18).

Reassigned corpora remove the stored shape while keeping the run structure: a kind's shape changes from pass to pass, so only the previous sentence about the kind shows which shape comes next. Trained on them in grouped order, models tested with trained nouns carried the shape forward after the same noun (Δ shape 10.01) and not after a different one (0.08). Their noun-specific index was 11.24 [10.44, 12.03] (Holm $p < .001$), close to the 11.4 that the corpus warrants. With novel nouns, which the models did not track, the noun-specific index was small (0.37 [0.24, 0.49], Holm $p = .001$). In random order, Reassigned corpora taught neither a preference for shape (0.21) nor a dependence on the noun (−0.07 with trained nouns), so the effect needed sentences about a kind to be close together.

So with trained nouns, the context's shape was carried forward only in Reassigned corpora, where the noun could not predict it, and there only after the same noun. This resembles the trade-off between in-context and in-weights learning that Chan et al. (2022) found with images, here in text with the content held fixed.

4.4 Fixed files memorise and change the result

Fixed-file models memorised their files: their loss was 0.08 to 0.11 nats per token on their own file but 0.66 to 1.26 on a new arrangement of the same sentences, while fresh-text models scored 0.28 to 0.55 on both (Appendix C). The arrangement effects changed with it. The grouping effect was about half the size (6.02 [5.24, 6.79] against 11.08), and in random order the index was 0.47 rather than −1.56, though not reliably above zero (Holm $p = .061$). Early in training, while their loss was near that of fresh-text models, fixed-file random-order models also gave −0.4 to −1.5 nats; their index rose above zero only late, as loss fell to the memorised level (exploratory). Every contrast measured in both regimes had the same direction. Their sizes differed, and the random-order index alone would have suggested no expectation, or a weak preference for shape, rather than an expectation that shape changes (fresh minus fixed, −2.03 [−2.42, −1.65]; exploratory).

Two cautions limit this result. Our fixed files were read 160 to 290 times, more than most artificial-language studies use, although BabyLM entries have trained for hundreds to thousands of epochs (Warstadt et al., 2023). And fixed against fresh text changes both the number of passes and the variety of windows, so we cannot say which matters. The result adds to evidence that repeated passes over fixed data change what is learned (Murty et al., 2023; Charton and Kempe, 2024): here they changed the measured effect of arrangement itself.

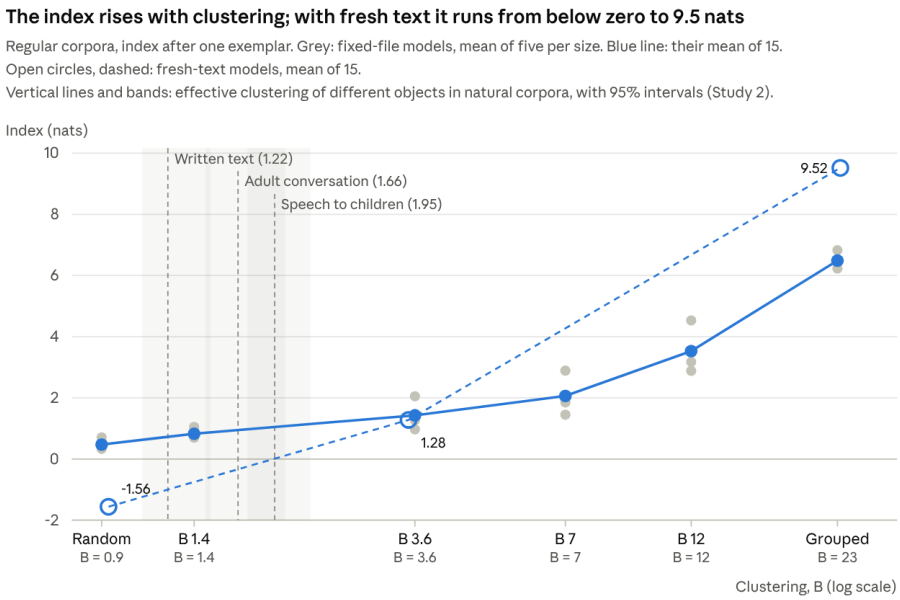


Figure 2: Index against clustering B in Regular corpora, novel nouns. Grey points and solid line: fixed-file models (mean of five per size; line, mean of 15). Open circles and dashed line: fresh-text models (mean of 15). Bands: effective clustering of different objects in natural corpora with 95% intervals (Section 5); the overlay does not predict what a model trained on them would learn.

5 Natural text and pretrained models

Natural corpora. With the same statistic, we measured how clustered mentions of 70 object nouns are in speech recorded around children under three (CHILDES; MacWhinney, 2000), adult conversation (CABank) and written text (the Pile; Gao et al., 2020). Labels were clustered in all three, most in speech to children (B = 3.57 [3.13, 4.30]; above written text, Holm $p < .001$; Table 4). But a repeated label need not bring a new object: in 500 hand-coded pairs, nearly half of the repetitions in speech to children returned to the same object, and a quarter brought a different one. Counting only different objects, effective clustering was 1.95, 1.66 and 1.22, which places all three corpora between random order and $B = 3.6$ on our scale, where the fixed-file index rose only from 0.47 to 1.43 nats (Figure 2). Speech to children differed more in density, with about 2.8 nearby different-object repetitions per 1,000 words against 0.1 to 0.2 elsewhere (exploratory; partly a result of our choice of nouns). Appendix J gives the method and checks.

Pretrained models. On English versions of the items, the index of Pythia models (Biderman et al., 2023) rose with size ($\rho = .92$, Holm $p = .003$); on items whose two words have the same number of tokens (exploratory), every model from 1B parameters was above zero (from 1.00 nats at 1B to 1.62 at 12B). It did not rise steadily over training: at every size it first fell to about −2 nats, as in our random-order models, and only larger models recovered. No Pythia model, nor Qwen2.5 72B or Llama 3.1 70B, preferred shape over texture, and evidence of a noun-tied expectation was weak and depended on wording (Appendix K). We did not arrange their data, so this is a comparison, not a test.

6 Discussion

Small transformers learned what the next sentence usually does. That much is expected of next-word prediction. What the controlled arrangement adds is which properties of the stream matter and how the stream interacts with what a model has stored. A regularity across items was taught by grouping only when a feature was fixed within kinds, and more selectively by nearby sentences about different objects than by the same object described again. With trained nouns, the models carried a shape forward from the context only when the noun could not predict it, and then only after the same noun, which resembles the trade-off between in-context and in-weights learning (Chan et al., 2022; Singh et al., 2023). We identify the training conditions that produce these behaviours, not their mechanism; carrying a word forward is the kind of be-

Registered test	Estimate [95% CI]	Holm p
Supported		
Grouped above random order	11.08 [9.57, 12.59]	<.001
$B = 3.5$ above random order	2.84 [1.41, 4.27]	.005
Grouped: Regular above No stable	13.50 [11.48, 15.52]	<.001
Grouped: Colour stable below No stable$^{\dagger\circ}$	2.76 [1.99, 3.52]	<.001
Index rises with $\log B$ (slope)$^{\dagger\circ}$	1.63 [1.46, 1.80]	<.001
Different-object above same-object runs	1.72 [0.54, 2.90]	.008
Noun-specific, Reassigned, trained nouns	11.24 [10.44, 12.03]	<.001
Noun-specific, Reassigned, novel nouns	0.37 [0.24, 0.49]	.001
above grouped Regular	0.25 [0.10, 0.39]	.005
Speech to children above written text, B	2.14 [1.55, 2.91]	<.001
Pythia index rises with size	$\rho = .92$	.003
Not supported		
Random-order index above zero$^{\dagger\circ}$	0.47	.061
512-token window raises index$^{\dagger\circ}$	0.03 [−0.44, 0.50]	.87
at $B = 3.6$ (reversed)$^{\dagger\circ}$	−0.63 [−0.90, −0.37]	1.00
Noun-specific, random order, novel nouns$^{\dagger\circ}$	0.005	.46
Pythia noun-specific index rises with size$^{\circ}$	$\rho = -.42$	1.00
Pythia noun-specific index above zero$^{\circ}$	−0.15	1.00
Pythia index rises over training	$\rho = -.34$	1.00
BabyLM baselines above zero	0.47; 0.37	.52; .12

Table 3: Main registered tests. Estimates are seed-level differences, or the index itself where no contrast is named; fresh text unless marked † (fixed files); $^{\circ}$ marks tests registered after related results were known, the others were registered before the related models were trained or scored, or before the transcripts were read. Units are nats except for B and ρ. Holm p is the test named in the registration, corrected within it: the seed-level test for fresh text, a mixed model or Welch test for fixed files, and for pretrained models the tests in Appendix K. Before a loader error was corrected, two fixed-file verdicts differed (Appendix A). Appendix A gives all 32 registered hypotheses.

haviour attributed to induction heads (Olsson et al., 2022), but we did not examine attention. People show related patterns: blocked study helps learners find what members of a category share (Carvalho and Goldstone, 2014), and varied exemplars taught toddlers more than near-identical ones (Perry et al., 2010). We note the parallel without taking our results as evidence about how children learn.

Two practical points follow for small models, with the caveat that we have not tested larger ones.

	Children	Adults	Written
B	3.57	2.33	1.43
Same-label pairs /1,000 words	10.1	0.9	1.0
Coded: same object	46%	25%	12%
Coded: different object	27%	25%	13%
Effective clustering	1.95	1.66	1.22

Table 4: Clustering of mentions of 70 object nouns within 100 words. Coded shares are from 500 hand-coded same-label pairs; effective clustering discounts B by the share of pairs that bring a different object (exploratory). Intervals are in Appendix J.

First, arrangement is a lever separate from content: to teach a regularity that holds across instances, place different instances of a kind near each other. Second, small-scale experiments on data should report how many passes they make over fixed text; arranging each pass afresh removes memorisation at no cost in content.

Limitations

The artificial language has 32 kinds, three dimensions and five frames that name shape more often than colour or texture; conclusions rest on comparisons between corpora that share the frames. The models are small (3.4M to 25.6M parameters), trained with one recipe and five seeds per condition, and we do not know how the effects scale. Colour stable corpora and half of the levels of clustering exist only as fixed files, which their models memorised. Fixed and fresh text differ in both the number of passes and the variety of windows, and our fixed files were read more often than in most published studies. Novel test nouns had untrained input embeddings, so the models could not recognise them as the same kind. Trained-noun items never offer the noun's own shape, so they show that the context's shape was ignored, not how the stored shape is used, and the comparison of trained with novel nouns is exploratory. The negative indices in random order and in No stable feature corpora are unexplained. The clustering statistic counts repeated labels, not referents, and effective clustering rests on 500 pairs coded by one person, with a language model as a consistency check. All corpora are English. The registrations were made in stages, several after earlier results were known, and a loader error, corrected before the final analyses, had changed two verdicts. The analysis of pretrained models does not manipulate their training data.

Ethics statement

The corpus analysis is a secondary analysis of archived, de-identified transcripts (CHILDES, CABank) used under their terms; no data were collected from people, and no transcript excerpts are reproduced. Code, corpora, results and all trained models will be released; anonymised copies are provided as supplementary material.

Use of AI assistants

An AI assistant (Claude, Anthropic) was used to help design analyses, write code, draft and edit text and check citations, and as the second coder in a consistency check of the corpus coding. The authors are responsible for all content, including the accuracy of every citation.

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A Registration record

In the appendices and the registrations, Study 1 is the training experiments (Sections 3 and 4), Study 2 the natural corpora and Study 3 the pretrained models (Section 5).

All registrations are in one public project on the Open Science Framework (link withheld for review; an anonymised view-only link is in the supplementary material), and each states what was known when it was written. Table A1 lists them, Table A2 gives the label each registration uses for each row of Table A4, and Table A4 gives every registered hypothesis and its result.

Table A1: Registrations. Each was written before the models it concerns were trained or scored, or before the transcripts were read.

Registration	Written	Rows of Table A4	Known when it was written
Original	2 Mar 2026	Corpora, models and controls (Appendix D)	Nothing trained
Amendment 1	7 Oct 2026	1, 2, 3, 6, 16	Random-order models trained and scored on an earlier test, with the old loader; no grouped model trained
Amendment 2	7 Oct 2026	7 (texture added as a co-primary test)	No grouped model trained
Amendment 3	8 Oct 2026	23, 24, 25	Study 1 results for rows 1 to 3, 6, 7 and 16 (old loader); no pretrained model scored
Amendment 4	8 Oct 2026	20, 21	Studies 1 and 3 results; no transcript read
Amendment 5	9 Oct 2026	4, 5, 8, 9, 17, 18, 19, 22	All earlier results (Study 1 with the old loader); four reviews of a draft; five exploratory checks, including the confound of the noun with another
Amendment 6	11 Oct 2026	11, 12, 13	Amendment 5 results (old loader); a prediction for the new models fitted to them
Amendment 7	12 Oct 2026	10, 14, 15	All results, with corrected scores
Amendment 8	15 Oct 2026	26 to 32	All results, with corrected scores; three rounds of reviews, which noted that fixed-file models memorised their files and that the novel-noun test was uninformative; no fresh-text model trained

Table A2: Rows of Table A4 and the labels used in the registrations.

Row	Label	Row	Label	Row	Label	Row	Label
1	H4b	9	H13b	17	H14a	25	H8
2	H1b	10	H22	18	H14b	26	H23
3	H2b	11	H17	19	H15	27	H24
4	H11	12	H18	20	H9	28	H25
5	H12	13	H19	21	H10	29	H26
6	H5b colour	14	H20	22	H16	30	H27
7	H5b texture	15	H21	23	H6	31	H28
8	H13a	16	H3b	24	H7	32	H29

The loading correction. The model code ties the input embedding to the output layer. On the graphics card used for training (through DirectML) the tie was lost, so every Study 1 model was trained and saved with two separate matrices. The first version of the scoring code rebuilt each model with the tie in place before loading its weights, so the output matrix overwrote the input embedding. We found this on 12 October 2026, fixed the loader, and rescored all 315 Study 1 models then trained, 461 intermediate checkpoints of the random and grouped Regular models and the trained-noun items, with the same items. A check on eight randomly chosen models confirmed the fix: mean loss on each model's own training

corpus was 0.07 to 0.16 nats per token with the corrected loader, close to the last logged training loss (0.08 to 0.23; dropout is off at test), and 0.58 to 1.97 with the old one. Studies 2 and 3 were not affected. The old scores are kept in the repository.

Table A3: Study 1 registered tests before and after the correction (shape-over-colour index, nats). Rows 10, 14 and 15 were registered and scored after the correction.

Row	Old loader	Corrected	Verdict before → after
1	1.36 vs 0.18	6.49 vs 0.47	Supported → supported
2	0.18	0.47	Not supported → not supported
3	0.18 vs 0.09	0.47 vs −0.25	Not supported → supported
4	difference 1.45	difference 6.93	Supported → supported
5	difference 1.17	difference 2.76	Supported → supported
6	1.36 vs 0.80	6.49 vs 2.12	Supported → supported
7	1.10 vs −0.04	5.36 vs 0.46	Supported → supported
8	slope 0.29	slope 1.63	Supported → supported
9	0.24 vs 0.18	1.43 vs 0.47	Not supported → supported
11	0.21 vs 0.18	0.50 vs 0.47	Not supported → not supported
12	0.29 vs 0.24	0.80 vs 1.43	Not supported → not supported (reversed)
13	0.21 vs −0.08	0.50 vs 0.10	Supported → supported
16	difference 0.0007	difference 0.005	Equivalent to zero → equivalent to zero

Table A4
All registered hypotheses and results; registered secondary analyses are reported in the text. Rows 1 to 16 concern fixed-file models and rows 26 to 32 fresh-text models. Seed-level tests average the three sizes within each of five seeds (t-based 95% intervals, two-sided; tests one-sided unless marked, 4 degrees of freedom), so a one-sided test can be below .05 while the two-sided interval includes zero (row 13). The registered p is the test named in the registration, one-sided unless marked, Holm-corrected within its registration; for rows 26 to 32 it is the seed-level test, corrected within rows 26 to 29 and within rows 30 to 32. Corpus intervals resample sources and then documents; Pythia and BabyLM intervals resample nouns.

#	Prediction	Result (nats)	Seed-level difference [95% CI]	Seed-level p	Registered p	Verdict
	1. Does grouping teach the fixed feature?					
1	Grouped order gives a higher index than random order	6.49 vs 0.47	6.02 [5.24, 6.79]	$< .001$	$< .001$	Supported
2	Random order gives an index above zero	0.47	0.47 [−0.61, 1.56]	.15	.061	Not supported
3	Random order: Regular above Mixed-slot	0.47 vs −0.25	0.73 [−0.69, 2.15]	.11	.032	Supported by the registered test, not at the seed level
4	Grouped: Regular above No stable feature	6.49 vs −0.44	6.93 [5.47, 8.40]	$< .001$	$< .001$	Supported
5	Grouped: Colour stable below No stable feature	−3.20 vs −0.44	2.76 [1.99, 3.52]	$< .001$	$< .001$	Supported
6	Grouped: Regular above Mixed-slot, shape over colour	6.49 vs 2.12	4.37 [1.98, 6.76]	.004	$< .001$	Supported
7	Grouped: Regular above Mixed-slot, shape over texture	5.36 vs 0.46	4.89 [2.69, 7.10]	.002	$< .001$	Supported
8	The index rises with log B across six orders	Slope 1.63 (SE 0.10)	1.63 [1.46, 1.80]	$< .001$	$< .001$	Supported
9	B = 3.6 above random order	1.43 vs 0.47	0.96 [0.63, 1.28]	$< .001$	$< .001$	Supported
10	The same, Objects corpora, different-object runs	1.18 vs 0.22	0.96 [0.37, 1.54]	.005	$< .001$	Supported
26	Fresh text: grouped order gives a higher index than random order	9.52 vs −1.56	11.08 [9.57, 12.59]	$< .001$	$< .001$	Supported
27	Fresh text: B = 3.5 above random order	1.28 vs −1.56	2.84 [1.41, 4.27]	.003	.005	Supported
28	Fresh text, grouped: Regular above No stable feature	9.52 vs −3.98	13.50 [11.48, 15.52]	$< .001$	$< .001$	Supported
	2. What kind of grouping helps?					
11	512-token window above 128, random order	0.50 vs 0.47	0.03 [−0.44, 0.50]	.44	.87	Not supported
12	512-token window above 128, B = 3.6	0.80 vs 1.43	−0.63 [−0.90, −0.37]	.003 (two-sided)	1.00	Not supported; reliably reversed
13	512-token window: Regular above No stable feature	0.50 vs 0.10	0.40 [−0.04, 0.85]	.032	.045	Supported; weak
14	Grouped: different-object above same-object runs	5.65 vs 2.99	2.66 [1.57, 3.75]	.001	$< .001$	Supported
15	B = 3.6: different-object above same-object runs	1.18 vs 0.45	0.73 [0.15, 1.32]	.013	$< .001$	Supported
29	Fresh text, grouped: different-object above same-object runs	7.60 vs 5.87	1.72 [0.54, 2.90]	.008	.008	Supported
	3. Is the expectation tied to the noun?					
16	Fixed files, random order: index higher after the same noun (novel nouns)	Difference 0.005; equivalent to zero (p = .001), but the novel nouns were untrained (Section 4.3)			.46	Not supported
30	Shape reassigned, grouped: noun-specific index above zero, novel nouns	0.37	0.37 [0.24, 0.49]	$< .001$	.001	Supported

#	Prediction	Result (nats)	Seed-level difference [95% CI]	Seed-level p	Registered p	Verdict
31	The same, above grouped Regular (fresh text)	0.37 vs 0.12	0.25 [0.10, 0.39]	.005	.005	Supported
32	Shape reassigned, grouped: noun-specific index above zero, nouns from training	11.24	11.24 [10.44, 12.03]	$< .001$	$< .001$	Supported
17	Pythia: noun-specific index rises with size (wording matched)	$\rho = -.42$			1.00	Not supported
18	Pythia 410M to 12B: noun-specific index above zero (wording matched)	−0.15 (SE 0.12)			1.00	Not supported
19	Pythia: 410M above 160M across ten training seeds 4. What do natural input and pretrained models show?	0.15 vs −0.15			.012	Supported
20	Speech to children: B above 1	3.57 [3.13, 4.30]			$< .001$	Supported
21	Speech to children above written text	3.57 vs 1.43; difference 2.14 [1.55, 2.91]			$< .001$	Supported
22	Speech to children differs from adult conversation (two-sided)	3.57 vs 2.33; difference 1.24 [0.51, 2.03]			$< .001$	Supported; higher
23	Pythia: index rises with size	$\rho = .92$			.003	Supported
24	Pythia: index rises over training	Mean within-size $\rho = -.34$			1.00	Not supported
25	BabyLM baselines (10M, 100M words): index above zero	0.47 [−1.05, 1.93]; 0.37 [−0.04, 0.73]			.52; .12	Not supported

Fresh text and shape reassignment. Rows 26 to 32 were registered on 15 October 2026, after reviews of the manuscript had noted that fixed-file models memorised their files and that the test with novel nouns was uninformative, and before any fresh-text model was trained. The corpora were built, and their checksums registered, before training. Amendment 8 names the conditions by their folder names (for example reassigned_grouped).

B Deviations from the registrations

1. Two registered analyses are not reported here: the Bayesian reference model of Amendment 1, which is in the repository, and the prediction of Amendment 6 for the 512-token models, which had been fitted to scores later corrected (Appendix A).

2. Added after registration: the seed-level tests for rows 1 to 3 and 6 to 8, the seed-level intervals, and intervals for the differences between natural corpora (from the registered two-stage bootstrap). Effective clustering was computed before it was registered as the scale for Figure 2 and is reported as exploratory (Table J1).

3. The two models of about 70 billion parameters were scored with Apple's MLX library, because PyTorch crashed the machine; on smaller models the two differed by 0.003 to 0.010 nats on average.

4. In the hand coding of Study 2, a rule for labels not used for a physical object was added during coding and applied by both coders, and one pair of the reliability subset was discussed with an AI assistant and left out of the agreement check.

Smaller technical changes (software versions, download addresses, a stale file in one public model repository) are listed in the repository.

C Model and training settings

Table C1: AdamW (betas 0.9 and 0.95, weight decay 0.1), linear warm-up then cosine decay, dropout 0.1, untied input and output embeddings. Fixed-file models read windows starting at random positions in the corpus; fresh-text models read non-overlapping windows, each at most once, in random order (the 3.4M and 25.6M models used about 60% of the windows available, the 11.0M models 94 to 96%). Byte-pair tokeniser with 351 entries for the Regular corpora and 314 to 339 for the other corpus families; every artificial word is one token. Seeds 42, 123, 456, 789 and 1001 each set the corpus, the initial weights and the batch order. Checkpoints were saved every 500 steps. Training ran on one Windows PC through DirectML.

Setting	3.4M	11.0M	25.6M
Layers, heads, width	4, 4, 256	6, 8, 384	8, 8, 512
Feed-forward width	1,024	1,536	2,048
Steps	5,000	8,000	10,000
Batch (128-token windows)	32	32	16
Batch (512-token windows)	8	8	4
Peak learning rate	3×10^{-4}	2×10^{-4}	1×10^{-4}
Warm-up steps	200	400	500

Table C2: Memorisation. Mean next-token loss (nats per token) on 300 windows from each model's own training text and from a new arrangement of the same sentences (100 further passes, 27 for the Objects corpora, never used in training); 15 models per cell. At the seed level, the gap between the two losses was larger for fixed-file than for fresh-text models by 0.57 to 1.16 nats, with every interval excluding zero.

Corpus	Fixed file: own file	Fixed file: new arrangement	Fresh text: own text	Fresh text: new arrangement
Regular, random	0.11	1.25	0.55	0.55
Regular, B 3.6 (fresh: 3.5)	0.10	1.26	0.54	0.54
Regular, grouped	0.09	0.85	0.35	0.35
No stable feature, grouped	0.09	1.00	0.39	0.39
Objects, same-object runs, grouped	0.08	0.66	0.28	0.28
Objects, different-object runs, grouped	0.09	0.76	0.32	0.32

D Further content controls

Six further corpora, each trained in random order only, change one thing about Regular. **No kind label**: the nouns are removed, with no other marker of the kind. **Paraphrased**: the nouns are removed and replaced by randomly chosen category markers. **Weak label**: nouns appear in 25% of sentences and category markers in the rest. **Frequency matched**: the same feature words as often as in Regular, with the fixed feature of each kind drawn at random from shape, texture or none. **Added noise**: Regular with 7% of exemplars given a wrong fixed feature. **Feature swap**: 16 kinds keep their shape and 16 keep their texture.

Table D1: Further controls, random order, 15 models each. In random order every index is small, so these corpora say little about what grouping would do with them.

Corpus	Shape over colour, mean (SD)	Shape over texture	Noun-specific index
Regular (for comparison)	0.47 (0.90)	0.51	0.005
No kind label	0.06 (0.49)	0.44	−0.004
Paraphrased	0.41 (0.44)	0.60	−0.025
Weak label	0.09 (0.50)	−0.52	0.008
Frequency matched	−0.22 (0.63)	−0.40	−0.033
Added noise	−0.43 (0.67)	0.44	−0.038
Feature swap	0.69 (0.77)	0.60	−0.017

E Robustness

Table E1
p values for the Study 1 rows of Table A4 under paired tests at the seed level and over the 15 matched models (one-sided unless marked).

Row	Seed level (5 seeds)	Matched models (15)
1	$< .001$	$< .001$
2	.15	.031
3	.11	.014
4	$< .001$	$< .001$
5	$< .001$	$< .001$
6	.004	$< .001$
7	.002	$< .001$
8	$< .001$	
9	$< .001$	$< .001$
10	.005	$< .001$
11	.44	.44
12	.003 (two-sided)	.003 (two-sided)
13	.032	.015
14	.001	$< .001$
15	.013	$< .001$
26	$< .001$	$< .001$
27	.003	$< .001$
28	$< .001$	$< .001$
29	.008	$< .001$
30	$< .001$	$< .001$
31	.005	.010
32	$< .001$	$< .001$

Table E2
Shape-over-colour index by model size (one exemplar), and with two or four exemplars in the context that share their shape (all sizes). Rows marked fresh are fresh-text models; the rest are fixed-file models. In grouped Regular models, two or four exemplars lowered the index: a run of a kind has four sentences, so a fifth sentence about the same kind is rare (Section 4.1). In grouped No stable feature and Colour stable models, where shape varies within a kind, more exemplars sharing a shape lowered the index (Δ shape −0.64, −1.93 and −3.42 with one, two and four exemplars in grouped No stable feature models), for which we have no tested account; in random-order Regular models, too, the index was lower with two or four exemplars than with one.

Condition	3.4M	11.0M	25.6M	Two exemplars	Four exemplars
Regular, random, fresh	−1.00	−0.87	−2.81	−1.83	−2.19
Regular, B 3.5, fresh	1.37	2.27	0.21	2.05	−0.53
Regular, grouped, fresh	7.43	12.57	8.56	7.88	4.24
No stable feature, grouped, fresh	−1.21	−4.80	−5.92	−5.41	−7.24
Objects, same-object, grouped, fresh	8.21	4.60	4.81	3.86	0.62
Objects, different-object, grouped, fresh	9.96	6.29	6.54	5.78	2.13
Reassigned, random, fresh	−0.06	0.33	0.35	0.59	0.23
Reassigned, grouped, fresh	6.52	14.05	8.85	9.62	8.26
Regular, random	0.33	0.71	0.38	−0.64	−0.10
Regular, B 1.4	0.73	1.05	0.70	−0.01	0.32
Regular, B 3.6	0.97	2.05	1.27	0.60	1.06
Regular, B 7	1.85	2.89	1.45	1.23	1.15
Regular, B 12	3.18	4.53	2.88	2.85	2.43
Regular, grouped	6.41	6.83	6.23	5.24	4.30
Mixed-slot, random	−0.11	−0.45	−0.21	0.29	0.29
Mixed-slot, grouped	2.24	2.35	1.76	2.60	2.04
No stable feature, random	−0.10	−0.33	−0.12	−0.17	−0.44
No stable feature, grouped	−0.41	−0.28	−0.63	−1.77	−3.26
Colour stable, random	−0.04	−0.70	−0.51	−0.76	−0.76
Colour stable, grouped	−2.11	−3.72	−3.77	−4.50	−5.34
Regular, random, 512	0.30	0.51	0.70	0.51	0.32
Regular, B 3.6, 512	0.31	1.02	1.06	0.93	0.82
No stable feature, random, 512	0.21	0.04	0.05	−0.10	−0.23
Objects, random	0.36	0.44	−0.13	0.38	1.02
Objects, same-object, B 3.6	0.22	0.98	0.14	0.32	1.22
Objects, different-object, B 3.6	0.83	1.75	0.96	1.49	2.18
Objects, same-object, grouped	3.00	3.33	2.62	1.30	1.34
Objects, different-object, grouped	6.18	5.43	5.34	4.54	4.14

Noun-specific index. At the seed level, the noun-specific index was equivalent to zero (two one-sided tests, ±0.2 nats) in 25 of the 26 fixed-file conditions; the exception was same-object runs at B = 3.6 (−0.11, p = .058). No condition exceeded 0.15 nats in absolute value. These nulls are uninformative, because the models did not track novel nouns (Section 4.3). As a further check, for the 60 models in the random and grouped Regular and grouped Objects conditions we compared the log probability of the same novel noun after the context and the word A with the mean over the other seven novel nouns. The difference was no more than 0.032 nats in absolute value in every model, and positive on about half the items. Because the novel nouns were never training targets, this check also depends on output weights that training only pushed down, so it is weaker evidence than the trained-noun contrast in Section 3.5.

Correction across all tests. Holm correction was applied within each registered family of tests, as registered. Every row of Table A4 with a Holm-adjusted p below .001 has an unadjusted p below .001, and so stays below .05 under Holm correction across all 32 tests.

What the corpora reward. For every pair of consecutive sentences in each training corpus (five seeds pooled), Table E3 gives the share about the same kind and how often the second sentence repeats the first one's shape or colour. The noun-blind reference index is D(shape) − D(colour), where D = ln p − ln((1 − p)/9) and p is the share of pairs that repeat that feature, with ten values per dimension; it is what a learner that predicts each feature from the previous sentence alone would show on a one-exemplar item. The noun-aware reference computes the same index separately for pairs about the same kind and pairs about different kinds (add-0.5 smoothing).

Table E3: What each training corpus says about the next sentence, and the models' shape-over-colour index for comparison. Rows marked fresh describe fresh text, counted on 100 held-out passes arranged by the same rule (27 for the Objects corpora); the rest describe the fixed files. All values in nats except the shares. Code is in the repository.

Corpus	Next sentence same kind	Shape repeats, same kind	Shape repeats, other kind	Colour repeats, same kind	Noun-blind reference	Noun-aware, same kind	Noun-aware, other kind	Model index
Regular, random, fresh	0.03	1.00	0.07	0.10	−0.01	11.7	−0.35	−1.56
Regular, B 3.5, fresh	0.11	1.00	0.07	0.09	0.65	13.1	−0.37	1.28
Regular, grouped, fresh	0.72	1.00	0.07	0.10	3.25	14.9	−0.38	9.52
No stable feature, grouped, fresh	0.72	0.10	0.10	0.10	0.02	0.0	−0.01	−3.98
Objects, same-object, grouped, fresh	0.76	1.00	0.07	0.99	0.39	8.4	−0.35	5.87
Objects, different-object, grouped, fresh	0.76	1.00	0.07	0.10	3.43	14.9	−0.29	7.60
Reassigned, random, fresh	0.03	1.00	0.10	0.10	0.27	8.2	0.01	0.21
Reassigned, grouped, fresh	0.72	1.00	0.10	0.10	3.30	11.4	−0.02	9.80
Regular, random	0.03	1.00	0.07	0.10	−0.01	10.3	−0.35	0.47
Regular, B 1.4	0.04	1.00	0.07	0.10	0.13	10.8	−0.35	0.83
Regular, B 3.6	0.11	1.00	0.07	0.10	0.70	11.7	−0.31	1.43
Regular, B 7	0.22	1.00	0.07	0.10	1.23	12.4	−0.37	2.06
Regular, B 12	0.38	1.00	0.07	0.09	1.91	13.0	−0.40	3.53
Regular, grouped	0.72	1.00	0.07	0.10	3.25	13.5	−0.33	6.49
No stable feature, grouped	0.72	0.10	0.10	0.10	0.03	0.0	0.06	−0.44
Colour stable, grouped	0.72	0.10	0.10	1.00	−3.26	−12.5	0.38	−3.20
Objects, same-object, B 3.6	0.11	1.00	0.07	0.70	0.19	8.6	−0.35	0.45
Objects, different-object, B 3.6	0.11	1.00	0.07	0.11	0.74	11.6	−0.25	1.18
Objects, same-object, grouped	0.76	1.00	0.07	0.99	0.37	7.0	−0.39	2.99
Objects, different-object, grouped	0.76	1.00	0.07	0.10	3.39	13.5	−0.41	5.65

Table E4: Study 1 results for models trained on fixed files, in nats after one exemplar with the same noun; 15 models per row, 128-token windows unless marked.

Corpus	Order (B)	Δ shape	Δ colour	Shape over colour, mean (SD)	Models above zero	Shape over texture
Regular	Random (0.9)	0.50	0.03	0.47 (0.90)	10 of 15	0.51
Regular	Graded (1.4)	0.77	−0.05	0.83 (0.72)	12	0.81
Regular	Graded (3.6)	1.33	−0.10	1.43 (0.93)	15	1.28
Regular	Graded (7)	1.71	−0.36	2.06 (0.95)	15	1.95
Regular	Graded (12)	2.94	−0.59	3.53 (1.12)	15	3.09
Regular	Grouped (23)	5.36	−1.13	6.49 (1.49)	15	5.36
Mixed-slot	Random	−0.37	−0.12	−0.25 (0.72)	4	−0.54
Mixed-slot	Grouped	0.76	−1.36	2.12 (1.07)	14	0.46
No stable feature	Random	−0.01	0.17	−0.18 (0.39)	5	−0.09
No stable feature	Grouped	−0.64	−0.20	−0.44 (0.54)	4	−1.02
Colour stable	Random	−0.06	0.36	−0.42 (0.70)	4	−0.06
Colour stable	Grouped	−0.81	2.39	−3.20 (1.18)	0	−0.22
Regular	Random (0.9), 512-token window	0.37	−0.13	0.50 (0.62)	12	0.41
Regular	Graded (3.6), 512-token window	0.67	−0.13	0.80 (0.79)	11	0.67
No stable feature	Random, 512-token window	0.07	−0.03	0.10 (0.60)	9	−0.10
Objects	Random (1.0)	0.58	0.36	0.22 (0.43)	11	0.40
Objects	Same-object runs (3.6)	1.34	0.90	0.45 (0.62)	12	1.40
Objects	Different-object runs (3.6)	1.25	0.07	1.18 (0.67)	14	1.13
Objects	Same-object runs (24)	4.63	1.65	2.99 (1.05)	15	4.05
Objects	Different-object runs (24)	5.37	−0.28	5.65 (1.18)	15	5.07

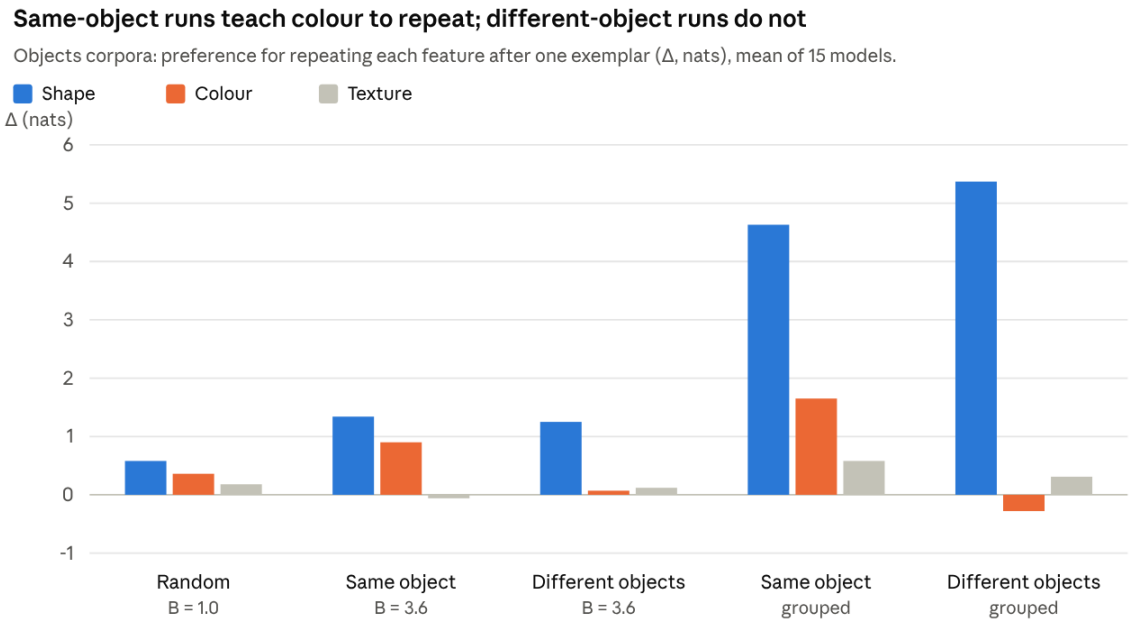


Figure E1: Preference for repeating each feature after one exemplar (Δ, nats; mean of 15 models per condition), in fixed-file models trained on the Objects corpora. Same-object runs raise the preference for repeating colour along with shape; different-object runs raise only shape.

F Trained nouns

All Study 1 models except those trained on the further content controls (Appendix D), 420 in all, were also scored on 960 items whose nouns occurred in training, in the format of Section 3.2: 16 context nouns, 10 draws of features, three probed dimensions and two continuation nouns (the same noun or another trained noun). No shape that is the fixed feature of either noun appears in an item.

Table F1: Trained nouns, selected conditions (all conditions in the repository). Rows marked fresh are fresh-text models; the rest are fixed-file models. Except after shape reassignment, the index on trained nouns is closer to zero than on novel nouns. Indices in nats; "models above zero" and p (two-sided one-sample t over 15 models) refer to the noun-specific index. For grouped Regular fixed-file models the registered one-sided test gives p = .008.

Condition	Index, same noun	Noun-specific index	Models above zero	p (two-sided)
Regular, random, fresh	−0.02	−0.09	7 of 15	.59
Regular, B 3.5, fresh	−0.13	−0.15	7	.19
Regular, grouped, fresh	2.00	1.04	10	.009
No stable feature, grouped, fresh	−3.75	−2.60	0	$< .001$
Objects, same-object, grouped, fresh	−0.55	0.05	8	.74
Objects, different-object, grouped, fresh	0.03	0.44	13	$< .001$
Reassigned, random, fresh	0.14	−0.07	5	.51
Reassigned, grouped, fresh	11.65	11.24	15	$< .001$
Regular, random	0.07	0.01	9	.97
Regular, B 3.6	0.36	0.12	10	.44
Regular, B 7	0.49	0.18	8	.29
Regular, B 12	1.07	0.06	8	.64
Regular, grouped	1.03	0.30	10	.015
Objects, same-object, grouped	−1.53	−1.06	0	$< .001$
Objects, different-object, grouped	0.33	0.14	10	.26

The comparison of novel and trained nouns in Table 2 (right) is exploratory and was added after the registrations; seed-level intervals for every fresh-text condition are produced by `scripts/recut_noun_store_check.py`.

G Materials for Studies 2 and 3

Object nouns (Study 2). ball, balloon, block, book, crayon, doll, puzzle, toy; airplane, bicycle, boat, bus, car, motorcycle, stroller, tractor, train, truck; apple, banana, carrot, cookie, cracker, egg, grape, raisin, sandwich; boot, button, coat, diaper, hat, jacket, mitten, necklace, pajamas, pants, shirt, shoe, sock, sweater, zipper; basket, blanket, bottle, bowl, box, broom, clock, cup, dish, fork, hammer, key, lamp, pencil, pillow, plate, purse, scissors, spoon, telephone, toothbrush, towel; bed, chair, couch, crib, table, television.

Coding rules (Study 2). Each pair was shown with the utterance before it, the stretch between the two mentions and the utterance after, without its source. *Same*: both mentions refer to the same physical object. *Different*: two or more distinct objects of the kind (another, other, numbers, plurals that count objects, descriptions that cannot fit one object). *Generic*: at least one mention refers to the kind in general. *Unclear*: the words do not settle it, including labels not used for a physical object. The second coder coded a random 25% of pairs (seed 2031). The full rubric and the second coder's prompt are in the repository.

Feature words (Study 3). Word list 1: shape *pointed, oval, circular, curved, rectangular, triangular, spherical, cylindrical*; colour *grey, purple, amber, violet, scarlet, crimson, maroon, turquoise*; texture *smooth, shiny, sticky, fuzzy, fluffy, furry, glossy, silky*. Word list 2: shape *oblong, hexagonal, conical, tapered, elliptical, tubular, cubic, rounded*; colour *magenta, lilac, teal, beige, indigo, lavender, ivory, coral*; texture *velvety, waxy, prickly, bumpy, gritty, coarse, slippery, rough*. Mean Zipf frequencies for list 1 are 3.75 (shape), 3.82 (colour) and 3.70 (texture).

Nouns (Study 3). Original items: snerg, twick, vulf, wazz, glorp, yeff, zull, kleb. Added sets: these and brell, sprock, fendle, plim, skeb, tulver, grisp, vosh.

Item counts (Study 3). Original items: 576 one-exemplar items (8 nouns, 12 draws, 3 dimensions, 2 continuation nouns) and 2,304 with two or four exemplars sharing either shape or colour, 2,880 in all. Each added set has 1,152 one-exemplar items.

Checkpoints (Study 3). Pythia 14M, 70M, 160M, 410M, 1B, 1.4B and 2.8B at steps 0, 4, 16, 64, 256, 1,000, 4,000, 16,000, 64,000 and 143,000 (2.8B without step 64,000); Pythia 6.9B and 12B at steps 64, 1,000, 4,000, 8,000, 12,000, 16,000, 32,000 and 143,000; Pythia 410M to 2.8B also at every 1,000 steps from 2,000 to 15,000 and at 18,000, 20,000, 24,000 and 32,000; nine further training seeds of Pythia 14M, 31M, 70M, 160M and 410M (and the original 31M model) at steps 4,000, 16,000, 64,000 and 143,000; the BabyLM GPT-2 baselines and Qwen2.5 72B and Llama 3.1 70B at their released weights. Exact revisions are listed in the repository.

H Data and code

All code, corpora, test items, model scores, transcript-level counts, hand codes (without transcript excerpts) and analysis outputs, and the final weights and training logs of the 510 trained Study 1 models, will be released on publication; anonymised copies of the code and results are in the supplementary material. A README gives the command that regenerates each table and figure, with MD5 checksums for the fixed inputs. The pretrained models are public at the revisions listed. CHILDES and CABank transcripts are available from TalkBank under its terms and are not redistributed. Study 2 is a secondary analysis of these archived, de-identified transcripts and of licensed telephone corpora, used under their terms; no data were collected from people for this paper, and no transcript excerpts are reproduced.

I Study 1: corpora, training and further results

The main text summarises the design. This appendix gives it in full, with results not reported there.

I.1 Corpora

Each corpus describes 32 kinds of object, each named by a novel noun (*wug*, *dax*, *blorp* and so on), with 14 exemplars per kind. Each exemplar has a shape, a colour and a texture, each a novel word from a list of ten per dimension, and is described by one sentence in one of five frames:

1. *A wug is a [shape] [colour] [texture] thing.*
2. *Look at the wug. It is [shape] and [colour].*
3. *Here is another wug. This one is [shape].*
4. *The wug I showed you is [shape], but a different [colour].*
5. *That is a wug because it is [shape].*

Shape is named in all five frames, colour in three and texture in one. The frames are the same in every corpus we compare, so they cannot by themselves explain a difference between corpora, although they may interact with what is manipulated. The 448 sentences of a corpus, each used once, make up one pass. Five corpora were generated for each condition, one per seed.

Content. The corpora differ in which feature, if any, stays fixed within a kind (Table eftab:1). In **Regular** corpora every kind keeps one shape across its 14 exemplars, while colour and texture vary. In **Mixed-slot** corpora each kind keeps one word from the shape list, but for about half the kinds that word appears in the texture slot, and colour varies; no single dimension is fixed across all kinds. In **No stable feature** corpora, built from Regular, each exemplar's shape is drawn at random from the list, so nothing is fixed within a kind. In **Colour stable** corpora, also built from Regular, shape is drawn in the same way and each kind keeps one colour instead. Frame 4 is unchanged, so in Colour stable corpora it names the kind's fixed colour after the words *a different*.

Objects. A further set of corpora separates two kinds of nearby repetition. In the Regular corpora, two sentences about a kind always describe different exemplars. In speech, nearby mentions often return to

the same object. The Objects corpora keep the Regular content but give each kind 13 objects and describe each object four times, once in each of frames 1, 2, 3 and 5. Frame 4 is left out because it states that the colour differs. A pass of these corpora has 1,664 sentences.

Order. The same sentences were arranged in several orders. In **random** order every pass is shuffled. In **grouped** order, within each pass, the sentences about each kind are shuffled, cut into runs of four, and the runs are shuffled across kinds (Figure 1). Every sentence occurs exactly as often as in random order. In **graded** orders each run is kept with a set probability and otherwise split into single sentences before shuffling, which gives intermediate levels of clustering. In the Objects corpora, a **same-object run** is the four descriptions of one object, and a **different-object run** is four different objects of one kind, one per frame. A Latin square over objects and frames ensures that every sentence is used once per pass in every order. Same-object and different-object runs were each arranged fully grouped and at an intermediate level.

Reassigned. In every corpus so far, a kind's fixed feature is fixed for the whole corpus, so the noun alone predicts it, and a model can predict the shape in a sentence from its noun without comparing the sentence with the one before. In Reassigned corpora, built from Regular, each kind receives a new shape, drawn at random from the ten, in every pass, and keeps it in all of its sentences in that pass. The next sentence about the same kind still repeats the shape, but the noun no longer says which shape it will be. These corpora were arranged in random and grouped order.

Fresh and fixed text. A model trained many times over on one short file can memorise it. Our main models were therefore trained on fresh text: 7,500 passes of a corpus (2,100 for the Objects corpora), each arranged with its own random draw by the rule of its order, 34 to 35 million tokens in all. Training read this text in non-overlapping 128-token windows, so no window was seen twice. A second set of models was trained in the common way, on one fixed file of 11,648 sentences (26 passes, or 7 for the Objects corpora; about 120,000 tokens), with windows starting at random positions, so that each position was seen 160 to 290 times. These fixed-file models memorised their files. Their loss was 0.08 to 0.11 nats per token on their own file but 0.66 to 1.26 on a new arrangement of the same sentences, while fresh-text models scored 0.28 to 0.55 on both (Table C2). Table I1 shows which conditions were trained on fresh text and which on a fixed file; where both exist, we report fresh text first. The fresh graded order, built by the same rule as the fixed one at B = 3.6, measured B = 3.5.

Clustering. We measure clustering with one statistic, B, here and in Study 2. Take the sequence of mentions of object nouns in a text, and every pair of consecutive mentions that are no more than 100 words apart. P is the share of these pairs that name the same kind. P_0 is the same share after the sentences (or utterances) of each document are shuffled, averaged over shuffles. B = P/P_0, pooled over documents. For example, in *Look at the dog. The dog is big. Where is the cup? Here is another dog.*, the consecutive pairs are dog–dog, dog–cup and cup–dog, so P = 1/3. B = 1 means no clustering beyond what the content predicts. Random-order Regular corpora have B = 0.9, slightly below 1, because each pass is shuffled separately: within a pass, a sentence's kind has 13 other members among 447 sentences, against 363 among 11,647 when the whole corpus is shuffled. The graded levels (Table eftab:1) were chosen after the clustering of natural input had been measured (Study 2): 1.4 and 3.6 match B in written text (1.43) and in speech to children (3.57). They were fixed from the corpora alone before any graded-order model was trained.

I.2 Models and training

We trained decoder-only transformers of three sizes: 3.4 million parameters (4 layers, 4 heads, width 256), 11.0 million (6 layers, 8 heads, width 384) and 25.6 million (8 layers, 8 heads, width 512). They were trained for 5,000, 8,000 and 10,000 steps with batches of 32, 32 and 16, peak learning rates of 3×10^{-4}, 2×10^{-4} and 1×10^{-4}, AdamW (betas 0.9 and 0.95, weight decay 0.1), linear warm-up, cosine decay and dropout of 0.1. Each training example is a 128-token window, which spans several sentences, so attention crosses sentence boundaries. The tokeniser for the Regular corpora has 351 entries (314 to 339 for the other corpus families), and every artificial word is a single token. A seed sets the corpus, the initial weights and the batch order (Table I1 gives the models per condition). For the window test, models were trained with 512-token windows and a quarter of the batch size, so that every step saw the same number

Table I1: Study 1 corpora. Every corpus describes 32 kinds. Fresh: 7,500 passes (2,100 for Objects), each arranged anew and read once in non-overlapping 128-token windows. Fixed: one file of 11,648 sentences (10,304 to 12,992 for three further controls: No kind label, Paraphrased and Weak label), read in 128-token windows at random positions unless marked. Each order of each corpus was trained at three sizes with five seeds (15 models). The further controls are reported in Appendix D (Table D1).

Corpus	What stays fixed within a kind	Orders (measured B)	Text	Models
Regular	Shape. Colour and texture vary.	Random (0.9); graded (3.5); grouped (23)	Fresh	45
No stable feature	Nothing	Grouped (23)	Fresh	15
Objects	Shape; each object described four times	Same-object runs (24); different-object runs (24)	Fresh	30
Reassigned	Shape within a pass; a new shape for each kind in every pass	Random (0.9); grouped (23)	Fresh	30
Regular	Shape	Random (0.9); graded (1.4, 3.6, 7, 12); grouped (23)	Fixed	90
Regular	Shape	Random (0.9); graded (3.6)	Fixed, 512-token window	30
Mixed-slot	One shape-list word, in the shape slot for about half the kinds and the texture slot for the rest	Random; grouped	Fixed	30
No stable feature	Nothing	Random; grouped; random with a 512-token window	Fixed	45
Colour stable	Colour. Shape varies.	Random; grouped	Fixed	30
Objects	Shape; each object described four times	Random (1.0); same-object runs (3.6, 24); different-object runs (3.6, 24)	Fixed	75
Further controls: no kind label, weak label, paraphrased, frequency matched, added noise, feature swap	As Regular, with one change each	Random	Fixed	90
Total				510

of tokens. All models have separate input and output embeddings, and the parameter counts include both; the model code ties the two, but the tie was lost on the training hardware (Appendix A). Table C1 lists every setting.

I.3 Measuring the expectation

Eight novel nouns that never occur in training (*snerg*, *twick* and so on) are used in test items. They are in the vocabulary, so each is one token, but their input embeddings were never trained; weight decay shrinks them by less than 10% over training, so they remain distinct random vectors. With the intended tie, the same vectors would have served as the novel nouns' output weights and would have changed in training; we cannot say whether the models would then have tracked these nouns. Novel nouns keep learned knowledge about a noun out of the test, but the test then depends on the model registering that the same untrained token recurs. We therefore also test with nouns from training, whose embeddings are trained (Appendix F). Each item gives the model one sentence in frame 1 describing an exemplar, followed by two continuations that differ in one word:

> Context: *A snerg is a krannish pulek skorli thing.* Repeat: *A snerg is a* ***krannish*** *temil tramel thing.* New: *A snerg is a* ***tova*** *temil tramel thing.*

In this item the probed dimension is shape. The repeat continuation keeps the context's shape, the new one does not, and both give new values for the other two dimensions. The model scores each continuation by the sum of the log probabilities of its tokens given the context, so the difference, Δ, includes the

changed word and its effect on the words after it. Δ is the model's preference for repeating the probed dimension, in nats. Trained models generally prefer words they have just seen, so Δ alone mixes the dimension with copying. Our index is the difference between dimensions:

$$\text{index} = \Delta_{\text{shape}} - \Delta_{\text{colour}}$$

Copying that affects shape and colour equally cancels. Copying that differs between dimensions does not, which is one reason every conclusion rests on comparisons between corpora tested in the same way, and why we report Δ for each dimension as well. The shape-over-texture index is defined in the same way. With one exemplar, nothing in the context shows which dimension is fixed, so differences between trained conditions come from training.

Every item is also given with a different novel noun in the continuations (*A yeff is a krannish temil tramel thing*). The **noun-specific index** is the index after the same noun minus the index after a different noun. A kind-tied expectation scores above zero on it; a noun-blind one scores zero.

With 25 draws of features per noun, three probed dimensions and two continuation nouns there are 1,200 one-exemplar items. A further 2,400 items with two or four exemplars test the use of evidence in the context (Table E2). Untrained models of the same architecture, with the Regular tokeniser, give an index of 0.02 nats (SD 0.07).

I.4 Statistics

Each model gives one index, its mean over items. Models of the three sizes trained with the same seed share a corpus, so the five seeds are the units of replication. We average the three sizes within each seed and compare conditions by paired t tests over the five seed means (4 degrees of freedom); the differences and 95% intervals given in the text are these seed-level estimates. Registered contrasts were tested one-sided, with Holm (1979) correction within each registered family. For fresh-text models the registered test was this seed-level test. For fixed-file models it was a linear mixed model (index $\sim$ condition + size, with a random intercept for seed) or, for some contrasts, a Welch t test over models; Table E1 gives the seed-level results beside them. Single conditions were compared with zero by one-sample t tests. Results reported with a Holm-corrected p, or called registered, were registered; all others are exploratory. We registered 0.2 nats as the smallest effect of interest and tested equivalence to zero against it with two one-sided tests (Lakens, 2017). This bound is small next to the effects reported here, so the equivalence tests are strict.

I.5 Further results

Mixed-slot corpora, trained as fixed files, keep a fixed word for every kind but no single fixed dimension. In grouped order these models preferred shape over colour (2.12), as their corpus supports, since colour is never fixed there, but showed no reliable preference between shape and texture (0.46), each of which is fixed for about half the kinds. Grouped Regular models were above them on both indices (differences 4.37 [1.98, 6.76] and 4.89 [2.69, 7.10], Holm $p < .001$). In random order, Regular models were above Mixed-slot models by the registered test (0.47 against −0.25, Holm p = .032), but not reliably at the seed level (3 of 5 seeds; p = .11).

Across the six orders of the fixed-file Regular corpora, the index rose with clustering at every step (Table E4, Figure 2). The registered test, a linear trend in the natural log of B, was supported (slope 1.63 nats per log unit, SE 0.10, Holm $p < .001$; seed level, 1.63 [1.46, 1.80]). The rise was steeper at high clustering: about 0.6 to 1.0 nats per log unit up to B = 7, and 2.7 and 4.6 nats per log unit over the two highest steps. At B = 3.6, fixed-file models were above random order (1.43 against 0.47; difference 0.96 [0.63, 1.28], Holm $p < .001$), as were fixed-file Objects models with different-object runs at the same level (1.18 against 0.22; difference 0.96 [0.37, 1.54], Holm $p < .001$). Fresh-text models agree with this curve at B = 3.5, but lie below it in random order and above it in grouped order (Figure 2).

In fixed-file models, under our matched-tokens recipe, a longer window did not raise the index (with tokens per step held fixed, the 512-token models also saw a quarter as many sequences per step, so the result applies to this recipe, not to window length alone; Table C1). With 512-token windows, which hold

about four times as many sentences, random-order Regular models gave 0.50 nats against 0.47 with 128 tokens (difference 0.03 [−0.44, 0.50], Holm p = .87), and the registered equivalence test did not show equivalence either (p = .15). At B = 3.6, the longer window gave less, 0.80 against 1.43 (difference −0.63 [−0.90, −0.37]; two-sided seed-level p = .003). With the longer window, Regular models were still above No stable feature models (0.50 against 0.10, Holm p = .045; seed level, 4 of 5 seeds, p = .032).

At matched clustering, runs of different objects gave a larger index than runs re-describing one object: 7.60 against 5.87 nats in grouped order (difference 1.72 [0.54, 2.90], Holm p = .008; higher in all five seeds). Fixed-file models showed the same (5.65 against 2.99; difference 2.66 [1.57, 3.75], Holm p $<$ .001), and also at B = 3.6 (1.18 against 0.45; difference 0.73 [0.15, 1.32], Holm p $<$ .001). Their same-object runs at B = 3.6 were neither reliably different from random order (0.45 against 0.22; difference 0.22 [−0.30, 0.74]) nor equivalent to it (registered secondary test).

J Study 2: clustering of natural input

Study 2 measures, with the statistic B of Study 1, how clustered mentions of object kinds are in three kinds of natural input: speech recorded around young children, conversation between adults, and the written text language models are trained on. It is descriptive. B is a shared statistic, not a calibrated dose: natural corpora differ from the Study 1 corpora in many ways besides clustering.

J.1 Method

Corpora. We used the North American English collection of CHILDES (MacWhinney, 2000), downloaded in full from TalkBank, kept every transcript whose target child was under 36 months, and took every utterance not spoken by that child. Most of this speech is addressed to the child, but some is addressed to others present; for brevity we call it speech to children. For adult conversation we used CABank: the Santa Barbara Corpus of Spoken American English (Du Bois and Englebretson, 2004), CallHome English (Canavan et al., 1997) and CallFriend English (Canavan and Zipperlen, 1996a,b). All but the Santa Barbara Corpus are telephone calls, in which the speakers share no objects in view. For written text we used the first 10,000 documents of the Pile (Gao et al., 2020), the training data of the Pythia models in Study 3, split into sentences. Documents of fewer than 400 words or with no mention of a target noun were excluded, leaving 2,116 transcripts from 35 CHILDES corpora, 245 CABank transcripts from four sources and 2,474 Pile documents from 21 subsets.

Kinds. We counted mentions of 70 count nouns from the MacArthur-Bates Communicative Development Inventory (Fenson et al., 2007) that name solid, inanimate objects, such as *ball*, *cup*, *shoe* and *truck*, the kinds for which children show the shape bias. Singular and plural count as one kind, and words often used as verbs or adjectives in speech to children, such as *brush* and *orange*, were left out. Because the nouns come from a list of words young children know, they are more frequent in speech to children than in the other corpora, by design.

Measure. B was computed as in Study 1, with the utterances or sentences of each document shuffled for P_0, within a window of 100 words, about the length of the 128-token training window of Study 1. Intervals resample sources (CHILDES corpora, CABank sources or Pile subsets) and then documents within them.

What repeated labels refer to. B counts repeated labels. It cannot tell two balls from one ball named twice, and Study 1 showed that the difference matters. One author coded 500 random same-label pairs (300 from speech to children, 100 from each of the others), with source labels withheld, as referring to the same object, to different objects of the kind, to the kind in general, or unclear; source could sometimes be guessed from the text. As a model-assisted consistency check, an instance of Claude (Anthropic), given only the coding rules and the pairs, coded a random quarter of them without access to the first coder's codes; the prompt and the codes are archived. Agreement on 124 pairs was 84% (Cohen's κ = 0.78). This is not a measure of agreement between human coders.

Effective clustering (exploratory). As a rough projection onto the Study 1 scale, we discounted each corpus's excess clustering by the share of coded pairs that referred to different objects, d, out of those that referred to particular objects: B_{eff} = 1 + (B − 1)d. A strict version takes d out of all coded pairs. Intervals

resample sources and documents for B and coded pairs for d. B_{eff} assumes that only different-object pairs inform the learner. Study 1 shows that same-object repetition is not neutral, so B_{eff} is a heuristic, not a corrected estimate.

J.2 Results

Object labels were clustered in every corpus, most of all in speech to children (Table J1). In speech to children B was well above 1 (registered test, Holm $p < .001$), above written text (difference 2.14 [1.55, 2.91], Holm $p < .001$) and above adult conversation (difference 1.24 [0.51, 2.03]; registered two-sided test, $p < .001$). The order of the three held in three registered secondary analyses: when each transcript was weighted equally, when every common noun was counted instead of our 70, and when documents were shuffled only within 500-word blocks, which keeps changes of topic. That last check lowered B in speech to children from 3.57 to 2.08, so part of the raw clustering reflects topic.

Table J1: Clustering of mentions of 70 object nouns within 100 words. Intervals resample sources and then documents, and for B_{eff} also coded pairs. B_{eff} is exploratory.

Measure	Speech to children	Adult conversation	Written text
B [95% CI]	3.57 [3.13, 4.30]	2.33 [2.02, 2.93]	1.43 [1.22, 1.86]
B, each transcript weighted equally	2.91	2.27	1.31
B, shuffling only within 500-word blocks	2.08	1.28	1.16
B, all common nouns	9.65	5.73	2.07
Same-label pairs per 1,000 words	10.1	0.9	1.0
Coded pairs: different objects	27% (82 of 300)	25% (25 of 100)	13% (13 of 100)
Coded pairs: same object	46% (139)	25% (25)	12% (12)
Coded pairs: kind in general or unclear	26%	50%	75%
B_{eff} [95% CI]	1.95 [1.73, 2.28]	1.66 [1.44, 2.04]	1.22 [1.09, 1.47]
B_{eff}, strict	1.70 [1.53, 1.96]	1.33 [1.21, 1.54]	1.06 [1.02, 1.13]

Nearly half of the repeated labels in speech to children referred again to the same object, and about a quarter brought a different object, a share similar to adult conversation. Most repeated labels in written text did not refer to a physical object, or referred to the kind in general. Effective clustering of different objects was therefore lower than B in every corpus. Speech to children was above written text on this measure (difference 0.73 [0.41, 1.09]) but not reliably above adult conversation (0.29 [−0.14, 0.69]).

Placed on the Study 1 scale by effective clustering, as registered, written text lies between random order and the graded order at B = 1.4, and adult conversation and speech to children lie between the graded orders at B = 1.4 and B = 3.6 (Figure 2). Over that range the index of fixed-file models rose only from 0.47 to 1.43 nats; fresh-text models were trained only near its ends, where they gave −1.56 in random order and 1.28 at B = 3.5. This does not say what a model trained on each corpus would learn.

The corpora differ more in how often such repetitions occur. Speech to children contained 10.1 same-label pairs per 1,000 words, against about 1 in the other two. Combining these rates with the coded shares, it contained about 2.8 nearby repetitions per 1,000 words that bring a different object of a kind, against about 0.2 in adult conversation and 0.1 in written text (exploratory). Part of this difference comes from our choice of nouns. B is a ratio and does not capture density, which Study 1 held fixed.

K Study 3: pretrained models

Study 3 asks whether models trained on large written corpora show the expectation, how it depends on size and training, and whether they tie it to the noun. It is a descriptive comparison: the arrangement of their training data is not manipulated.

K.1 Method

Items. The items follow the format of Study 1 in English. Each feature gets its own sentence, so English adjective order cannot favour one dimension, and the order of the three sentences is counterbalanced:

> Context: *Here is a snerg. It is cylindrical. It is purple. It is smooth.* Repeat: *Here is another snerg. It is **cylindrical**. It is turquoise. It is fluffy.* New: *Here is another snerg. It is **circular**. It*

is turquoise. It is fluffy.

Each dimension uses eight English words of similar frequency, and the nouns are novel words spelled with ordinary subword pieces (Appendix G). In the original items the same-noun continuation began *Here is another snerg* and the different-noun continuation *Here is a zull*, so the noun-specific index compared the noun and the word *another* together. Four further item sets with 16 nouns keep the wording of the two continuations the same: *a* in both (the registered primary set), *another* in both, a second frame, and a second word list.

Models. The Pythia models (Biderman et al., 2023; van der Wal et al., 2025) range from 14 million to 12 billion parameters and were trained on the same 300 billion tokens of the Pile in the same order. We scored their final checkpoints and checkpoints across training, and nine further training seeds of the five smallest sizes (van der Wal et al., 2025). We also scored the two GPT-2 baselines of the BabyLM Challenge (124 million parameters, trained on 10 and 100 million words; Charpentier et al., 2025), and Qwen2.5 72B (Qwen Team, 2024) and Llama 3.1 70B (Grattafiori et al., 2024) as descriptive reference points. Appendix G lists every checkpoint.

Tests. Trends with size are Spearman correlations across Pythia sizes; trends over training are tested by permuting checkpoints within size; intervals resample nouns. The pooled noun-specific test uses a mixed model with crossed random effects for nouns, items and feature words.

K.2 Results

The index rose with size but not steadily over training. At the end of training the shape-over-colour index rose with Pythia size (Spearman $\rho = .92$, Holm p = .003), from −2.44 nats at 14 million parameters to 1.17 at 12 billion (Figure K1). This raw index includes effects of the particular words. Untrained Pythia models give about 0.5 nats, because in some colour pairs the repeated and the new word take different numbers of tokens, and an untrained model pays roughly the same cost for every token. On the original items whose two words take the same number of tokens (all shape pairs and a quarter of the colour pairs; exploratory), untrained models gave −0.32 to 0.31 nats, the trend with size held ($\rho = .93$), and every Pythia model from 1B upwards gave an index above zero, from 1.00 [0.04, 1.82] at 1B to 1.62 [0.86, 2.19] at 12B. The trend with size held on every item set, including the second word list ($\rho = .83$ to .95; exploratory). The index did not rise steadily over training (mean within-size $\rho = -.34$). At every size, training first lowered it, to about −2 nats between steps 1,000 and 4,000, much as in the random-order models of Study 1, and only the larger models recovered. The BabyLM baselines did not show a reliable index (10 million words, 0.47 [−1.05, 1.93]; 100 million words, 0.37 [−0.04, 0.73]; Holm p = .52 and .12). On the items whose two words take the same number of tokens (exploratory), the 100-million-word baseline gave 1.16 [0.52, 1.82]. Pythia 160M, of similar size, gave a negative index (−1.22 [−1.66, −0.75]) even after 300 billion tokens, but the two families differ in more than their data, so this comparison does not separate data from size.

Weak, wording-dependent evidence of noun binding. On the original items the noun-specific index was positive in every Pythia model from 410 million parameters, but those items confounded the noun with *another*. With *a* in both continuations, the registered tests were not supported: the index did not rise with size ($\rho = -.42$, Holm p = 1.00), and pooled over Pythia 410M to 12B it was not above zero (−0.15 nats, SE 0.12, Holm p = 1.00). It was higher at 410M than at 160M across ten training seeds of each (0.15 against −0.15 nats, Holm p = .012), with 7 of the 10 seeds at 410M positive. With *another* in both continuations (a registered secondary analysis), it was above zero in four of the six Pythia models from 410M (0.28, 0.77, 0.30 and 0.17 nats at 410M, 1B, 2.8B and 6.9B), but not at 1.4B or 12B, and it was small and mixed in sign on the second frame and the second word list. Of the two reference models of about 70 billion parameters, Qwen2.5 gave a positive index with *a* in both continuations (0.56 [0.28, 0.84]) and Llama 3.1 did not (−0.16 [−0.36, 0.06]).

The components show why the wording matters. After *another*, which marks a new member of the kind, the larger models expected shape to repeat and colour to change more after the same noun than after a different one, and *another* also raised the raw index, to 0.49 to 1.08 nats in Pythia 1B to 12B (−0.34 to 0.46 with a in both). After *a*, the same noun may name the same object again, and then every feature

should repeat: colour repeated more after the same noun, and the noun-specific index fell.

Texture and context. No Pythia, Qwen or Llama model preferred shape over texture, so what the large models show is a shape-over-colour expectation, not a general shape bias (image classifiers trained on ImageNet likewise favour texture over shape; Geirhos et al., 2019). The BabyLM baseline trained on 10 million words did prefer shape over texture (1.17 [0.24, 2.34]). The two largest Pythia models tied texture to the noun more than shape. When two or four exemplars in the context shared a shape, every trained model followed the context, strongly so from 410 million parameters. What developed with size was the expectation after one exemplar, not the use of evidence in the context.

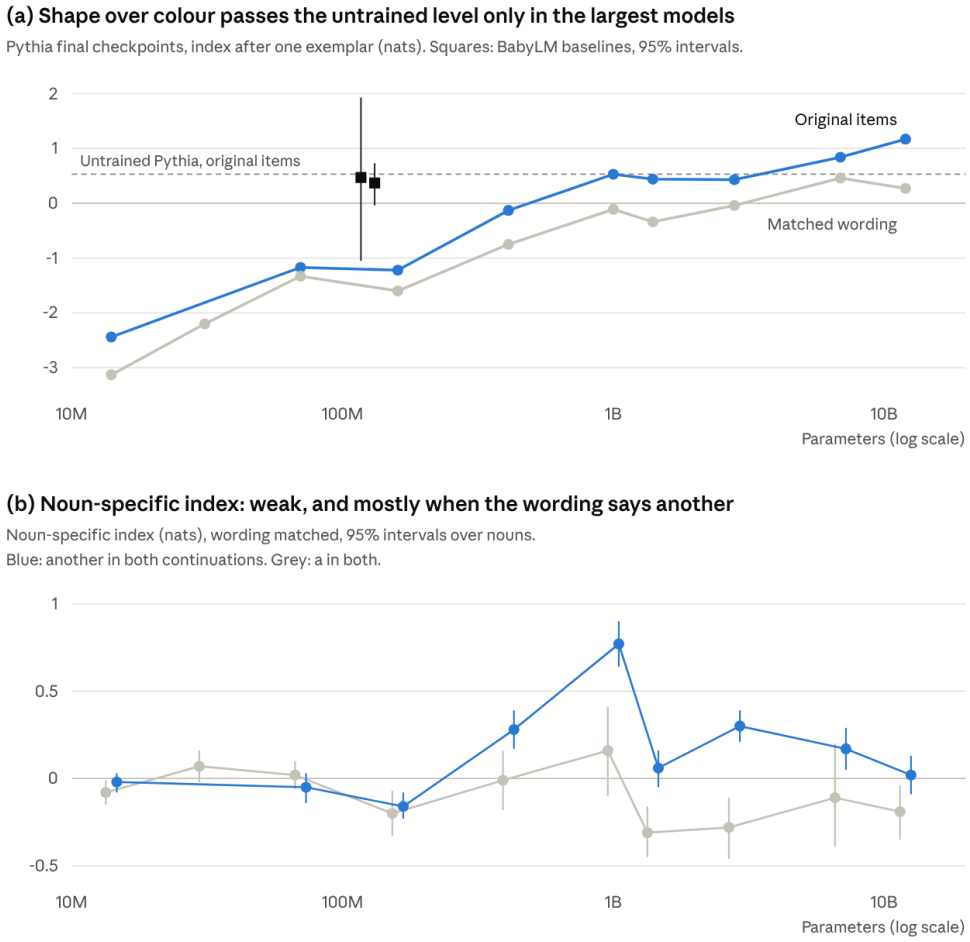


Figure K1: Pretrained models. (a) Shape-over-colour index (nats) at the end of training by Pythia size, on the original items and with matched wording (*a* in both continuations). Squares: BabyLM baselines with 95% intervals over nouns. Dashed line: untrained Pythia models on the original items (about 0.5 nats, from differences in token counts; Appendix K). (b) Noun-specific index with matched wording, with *a* or *another* in both continuations, with 95% intervals over nouns. In both panels, the 31M model was scored only with *a* in both continuations.